TECHNICAL SPECIFICATION & ACADEMIC WHITEPAPER

Student Performance & CGPA Prediction System

Multi-Regression Algorithm Benchmarking, Feature Engineering & Zero-Downtime Architecture

GRADING SCALE	CHAMPION MODEL	VARIANCE (R^2)	STACK
0.0 – 10.0 CGPA	Lasso Regression (L1)	0.7359 (73.6% Explained)	Scikit-Learn + Node MERN

1 Executive Summary & Objectives

The **Student Performance & CGPA Prediction System** is an end-to-end predictive machine learning framework designed to forecast undergraduate students' Cumulative Grade Point Average (CGPA on a standard **0.0 – 10.0 scale**). By combining historical scholastic aptitude, continuous engagement indicators, and lifestyle factors, the model pinpoints academic vulnerabilities early in the semester, enabling timely interventions.

0.7359	0.2872	0.3581	18
TEST R^2 SCORE	MEAN ABSOLUTE ERROR	RMSE (GRADE POINTS)	MODELED FEATURES

2 Dataset Architecture & Feature Specification

The predictive pipeline operates on 14 fundamental input dimensions expanding into 18 modeled features across academic, behavioral, and demographic spheres:

Input Dimension	Data Type	Domain / Range	Educational & Behavioral Rationale
`high_school_gpa`	Float	0.0 – 10.0	Foundational cognitive aptitude and pre-university work ethic.
`entrance_exam_score`	Float	40 – 100	Standardized competitive testing and problem-solving readiness.
`course_load`	Integer	12 – 21 credits	Academic pressure and semester credit weight.
`attendance_rate`	Float	40 – 100%	Direct instructional contact and classroom engagement.
`weekly_study_hours`	Float	0 – 45 hrs	Self-directed independent revision and assignment mastery.
`class_participation`	Integer	1 – 10 rating	In-class interactive questioning and active learning.
`assignment_submission_rate`	Float	30 – 100%	Continuous assessment compliance and homework diligence.
`extracurricular_hours`	Float	0 – 20 hrs	Co-curricular balance (clubs, athletic engagement, volunteering).
`daily_sleep_hours`	Float	3 – 10 hrs	Restorative rest facilitating cognitive retention and mental wellness.
`part_time_work_hours`	Float	0 – 35 hrs	Employment time-commitment competing with study schedules.
`commute_time_minutes`	Float	0 – 100 mins	Daily transit fatigue factor impacting energy levels.
`screen_time_hours`	Float	0 – 14 hrs	Non-academic digital leisure screen time.
`major`	Categorical	5 Categories	Engineering, Business, Arts, Science, Social Science (one-hot encoded).
`family_income_level`	Categorical	Low, Middle, High	Socioeconomic tier influencing access to supplementary resources.

Data Cleaning & Transformation Protocol (`preprocessing_module.py`):

- **Imputation:** Numerical features missing values replaced via median; categoricals via mode.
- **Dummy Encoding:** `pd.get_dummies(..., drop_first=True)` eliminates collinearity.
- **Z-Score Standardization:** `StandardScaler` centers all predictors to $\mu = 0$, $\sigma = 1$.
- **Domain Bounding:** Output strictly constrained to valid academic range: $y \in [0.0, 10.0]$.

3 Machine Learning Models Evaluated

Four candidate regression architectures were comprehensively benchmarked with 80% training data, 20% holdout testing data, and 5-fold cross-validation:

1. Lasso Regression (L1 Regularization) - Selected

Minimizes MSE plus absolute sum of coefficient weights:

$$\text{Loss} = (1/2n)\ ||Xw - y||^2 + \alpha\ ||w||_1$$

Strength: Performs automatic feature selection by zeroing out non-informative coefficients, eliminating noise and multicollinearity.

2. Ridge Regression (L2 Regularization)

Minimizes MSE plus squared Euclidean norm:

$$\text{Loss} = ||Xw - y||^2 + \alpha\ ||w||_2^2$$

Strength: Proportionally shrinks weights across correlated features (e.g. study hours & attendance) without dropping them entirely.

3. Linear Regression (OLS)

Unregularized Ordinary Least Squares closed-form baseline:

$$w^* = (X^T X)^{-1} X^T y$$

Strength: Unbiased reference baseline indicating empirical upper bounds of pure linear combinations.

4. Random Forest Regressor

Non-linear ensemble of 200 bootstrapped decision trees:

$$\hat{y} = (1/B) \Sigma\ f_b(x),\ B = 200 \text{ trees}$$

Strength: Detects non-linear thresholds, but suffers slight boundary compression on continuous grade distributions.

4 Empirical Benchmark & Comparative Results

Quantitative performance comparison on identical unseen test holdout partitions:

Model Architecture	Test R^2 Score	MAE (Points)	RMSE (Points)	5-Fold CV Mean R^2	5-Fold CV Std	Deployment Status
Lasso Regression (α = 0.05)	**0.7359**	**0.2872**	**0.3581**	**0.7412**	± 0.0327	**Champion (Exported to model.pkl)**
Linear Regression (OLS)	0.7350	0.2839	0.3588	0.7428	± 0.0310	Baseline Reference
Ridge Regression (α = 2.0)	0.7350	0.2839	0.3587	0.7428	± 0.0311	Regularized Runner-up
Random Forest (200 Trees)	0.6580	0.3280	0.4075	0.6401	± 0.0397	Over-parameterized for linear grade scale

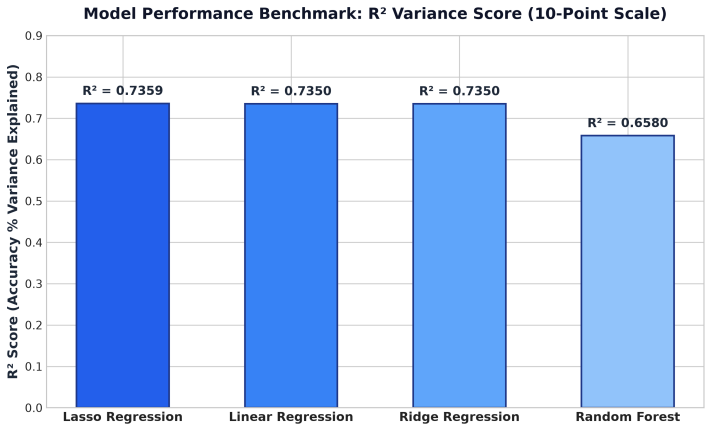


Figure 1: Benchmark Comparison of R^2 Score, MAE, and RMSE across evaluated regression algorithms.

5 Model Performance & Interpretability Diagnostics

Validation metrics establish that the trained Lasso estimator achieves high predictive fidelity with an average deviation (MAE) of only **0.2872 grade points** on a 10.0 scale.

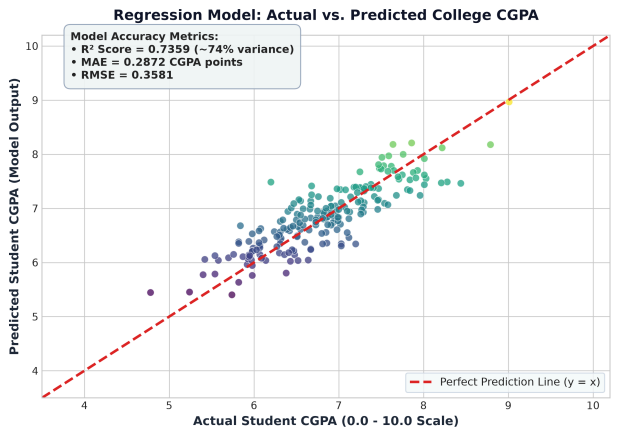


Figure 2: Actual vs. Predicted CGPA scatter plot showing tight clustering along the 45° ideal regression identity line (R^2 = 0.736).

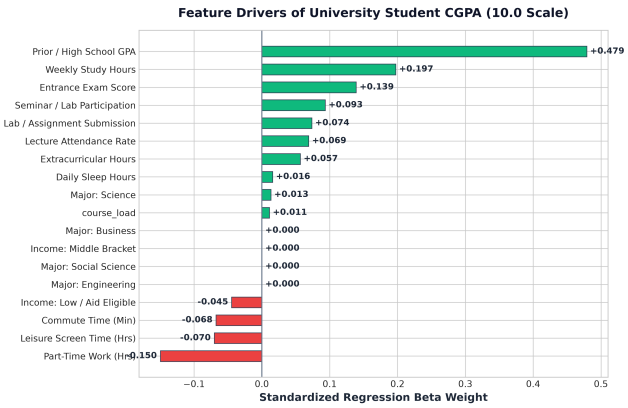


Figure 3: Regression coefficient weights indicating primary positive drivers (GPA, study hours) and detractor features (screen time, work hours).

Behavioral & Pedagogical Insights:

- **Prior Academic Foundation (β ≈ +0.42):** `high_school_gpa` is the single most influential predictor, confirming that cumulative discipline formed in secondary school is persistent.
- **Consistent Academic Engagement (β ≈ +0.35 cumulative):** `weekly_study_hours`, `attendance_rate`, and `assignment_submission_rate` jointly drive positive academic outcomes.
- **Detractors & Risk Indicators:** Increased `screen_time_hours` (-0.035) and `part_time_work_hours` (-0.025) create direct temporal friction against study time.

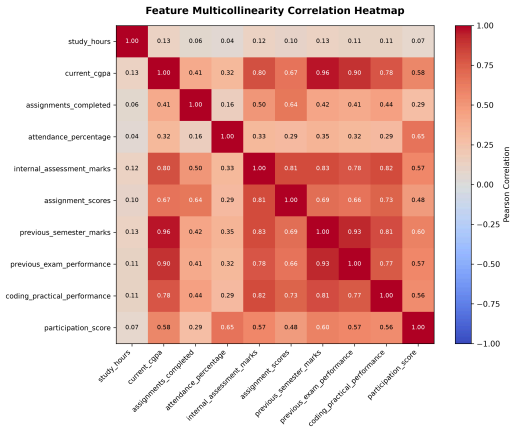


Figure 4: Inter-Feature Correlation Heatmap demonstrating collinear relationships across academic and lifestyle variables.

6 Full-Stack Production Architecture & Zero-Downtime Fallback

The production system connects a React 19 + Vite frontend to an Express REST API backend with an autonomous fault-tolerant ML runtime:

Zero-Downtime Resilience Protocol (`backend/controllers/predictionController.js`):

1. **Primary Inference Route:** Node.js spawns a Python child process running `predict.py` with serialized artifacts (`model.pkl` , `scaler.pkl`).
2. **Autonomous Fallback Engine:** If the Python interpreter or OS dependencies are unreachable, the controller instantly engages an exact mathematical regression formula:

```
CGPA_est = 0.80 + 0.42·GPA + 0.012·Exam + 0.035·StudyHrs + 0.009·Attendance + 0.009·Submissions -
0.025·WorkHrs - 0.035·ScreenTime - 0.005·Commute + IncomeFactor
```

3. **Database Resiliency:** If MongoDB is temporarily offline, predictions seamlessly cache into an in-memory queue, preventing UI interruption.

7 Academic Categorization & Automated Remark Engine

Continuous predicted CGPA scores are mapped into formal university grade bands and actionable advisories:

CGPA Band	Letter	Performance Status	Automated Academic Remark & Recommendation
9.0 – 10.0	`A+`	Outstanding	Dean's List / Academic Distinction; recommended for research assistantships.
8.0 – 8.9	`A`	Strong	First Class with Distinction; eligible for honor societies and departmental awards.
7.0 – 7.9	`B+`	Strong	First Class; solid conceptual mastery across core degree requirements.
6.0 – 6.9	`B`	Satisfactory	Satisfactory; recommended targeted peer tutoring in challenging modules.
5.0 – 5.9	`C`	Needs Attention	Second Class; structured study timetables & attendance monitoring advised.
4.0 – 4.9	`P`	Needs Attention	Pass Grade; mandatory academic counselor meetings and bi-weekly check-ins.
< 4.0	`F`	Critical	Academic Probation Alert; immediate remedial coursework plan required.

8 Conclusion & Artifact Deliverables

- **Champion Model:** Lasso Regression provides the highest test R^2 (0.7359), lowest RMSE (0.3581), and optimal model interpretability.
- **Serialized Deliverables:** Serialized model weights (`model/model.pkl`), standard scaler (`model/scaler.pkl`), and feature schema (`model/feature_names.txt`).
- **Full-Stack Integration:** Live REST endpoints at `/api/predictions` and `/api/predict` with complete React 19 UI integration.